

Embedded AI

Real-Time Fastener Detection on Raspberry Pi

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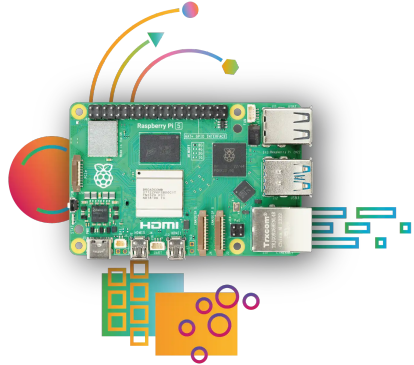
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Project goals

- ▶ Deploy real-time fastener detection on Raspberry Pi 4 & 5
- ▶ Explore edge-AI constraints (latency, power, memory)
- ▶ Evaluate state-of-the-art models and embedded performance



Workflow

- ▶ Set up Raspberry Pi 4/5:
 - ▶ Install OS
 - ▶ Install drivers for the camera - IDS SDK
- ▶ Record videos via the IDS camera, connected to the Raspberry Pi
- ▶ Annotate video frames via Roboflow framework
- ▶ Run training on the GPU
- ▶ Test real-time detection on Raspberry Pi

Camera (UI-3590CP-C-HQ Rev.2)

Sensor	
Resolution	18.10 Mpix (4912×3684)
Gain (master/RGB)	16× / 4×
Model	
Pixel clock range	96–432 MHz
Max frame rate (freerun/trigger)	21 / 21 fps
Exposure time (min–max)	0.013–998 ms
Power consumption	1.5–3.3 W
Image memory	128 MB
Environment & I/O	
Interface	USB 3.0 micro-B, screwable

iDS:

UI-3590CP-C-HQ Rev.2 (AB00606)

■ Discontinued
The model has been discontinued.



Roboflow

For annotation we used Roboflow. Roboflow is a platform for rapid image annotation, with APIs to upload images and export datasets in model-ready formats.



Raspberry Pi 5

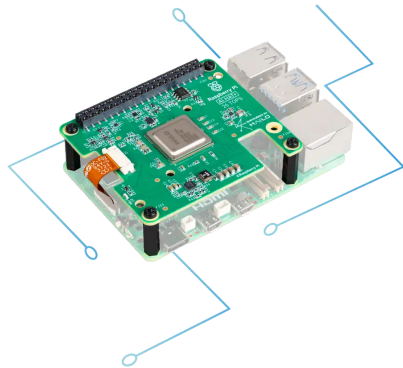
Compute

SoC	2.4 GHz quad-core 64-bit Arm Cortex-A76 (crypto ext.); 512 KB L2 per core, 2 MB shared L3
GPU	VideoCore VII; OpenGL ES 3.1, Vulkan 1.3

Memory and Storage

RAM	8 GB
microSD	64 GB

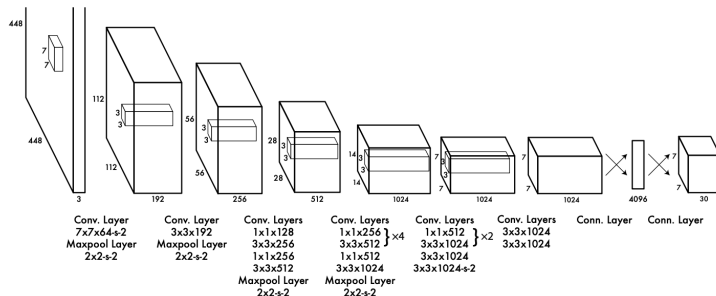
Hailo-8 accelerator offering 26 TOPS inferencing performance respectively



Training

YOLO remains the go-to family for the best speed–accuracy trade-off in object detection. We use the lightweight Small and Nano variants of YOLOv8 (2023) and YOLOv11 (2024) as baselines.

After YOLOv4 the naming became community-driven different teams publish their own “vX” lines. For example, Ultralytics maintains v5, v8, and v11;



Dataset

2131 Total Images

[View All Images →](#)



Dataset Split

TRAIN SET

93%

1983 Images

VALID SET

5%

98 Images

TEST SET

2%

50 Images

Preprocessing

Auto-Orient: Applied

Resize: Fit (black edges) in 960x960

Auto-Adjust Contrast: Using Contrast Stretching

Grayscale: Applied

Augmentations

Outputs per training example: 3

Flip: Horizontal, Vertical

90° Rotate: Clockwise, Counter-Clockwise, Upside Down

Hue: Between -24° and +24°

Saturation: Between -25% and +25%

Brightness: Between -22% and +22%

Exposure: Between -10% and +10%

Blur: Up to 2.5px

Noise: Up to 1.72% of pixels

Detection Metrics

For a predicted box p and ground-truth box g :

$$\text{IoU}(p, g) = \frac{|p \cap g|}{|p \cup g|}, P = \frac{\text{TP}}{\text{TP} + \text{FP}}, R = \frac{\text{TP}}{\text{TP} + \text{FN}}, \text{AP}_\tau = \int_0^1 p(r; \tau) dr,$$

$$\text{mAP}_{0.50} = \frac{1}{C} \sum_{c=1}^C \text{AP}_{c, 0.50},$$

$$\text{mAP}_{[0.50:0.95]} = \frac{1}{C} \sum_{c=1}^C \frac{1}{10} \sum_{k=0}^9 \text{AP}_{c, (0.50+0.05k)}.$$

- ▶ τ : IoU threshold for TP/FP (0.50 for mAP50; 0.50–0.95 for COCO mAP).
- ▶ AP is area under the precision–recall curve (confidence threshold is swept).
- ▶ C is the number of object classes we're evaluating over.

Results

Table: Validation summaries.

(a) YOLO8n

Class	Images	Instances	Box(P)	R	mAP ₅₀	mAP _{50–95}
all	98	363	0.927	0.951	0.964	0.733
machine_screw	25	123	0.931	0.967	0.989	0.625
miscellaneous	14	16	0.754	0.812	0.845	0.562
nut	16	41	0.988	1.000	0.995	0.777
washer	20	43	1.000	0.991	0.995	0.906
wood_screw	37	140	0.960	0.986	0.994	0.794

Speed per image (GPU): preprocess 4.5 ms; inference 6.0 ms; loss 0.0 ms; postprocess 2.5 ms.

Model summary (fused): 72 layers, 3,006,623 parameters, 0 gradients, 8.1 GFLOPs

Results

Table: Validation summaries.

(a) YOLO11s

Class	Images	Instances	Box(P)	R	mAP ₅₀	mAP _{50–95}
all	98	363	0.939	0.934	0.966	0.746
machine_screw	25	123	0.877	0.983	0.978	0.649
miscellaneous	14	16	0.899	0.688	0.867	0.565
nut	16	41	0.970	1.000	0.995	0.805
washer	20	43	0.973	1.000	0.995	0.909
wood_screw	37	140	0.978	1.000	0.995	0.802

Speed per image (GPU): preprocess 4.5 ms; inference 9.8 ms; loss 0.0 ms; postprocess 3.7 ms.

YOLO11s summary (fused): 100 layers, 9,414,735 parameters, 0 gradients, 21.3 GFLOPs

Results

Table: Validation summaries.

(a) YOLO11n

Class	Images	Instances	Box(P)	R	mAP ₅₀	mAP ₅₀₋₉₅
all	98	363	0.921	0.935	0.964	0.739
machine_screw	25	123	0.844	0.935	0.957	0.611
miscellaneous	14	16	0.922	0.741	0.888	0.608
nut	16	41	0.970	1.000	0.995	0.799
washer	20	43	0.996	1.000	0.995	0.902
wood_screw	37	140	0.873	1.000	0.983	0.773

Speed per image (GPU): preprocess 0.5 ms; inference 5.6 ms; loss 0.0 ms; postprocess 0.9 ms.

100 layers, 2,583,127 parameters, 0 gradients, 6.3 GFLOPs.

Results

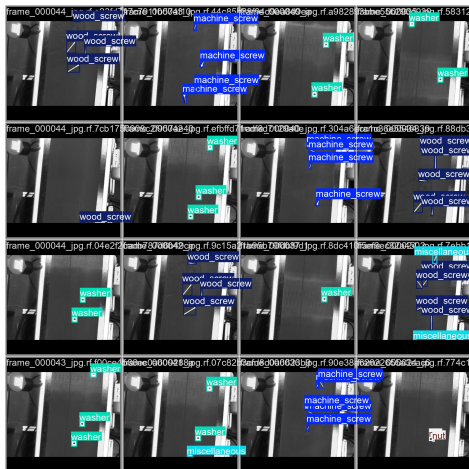


Figure: Ground truth: yolov8n

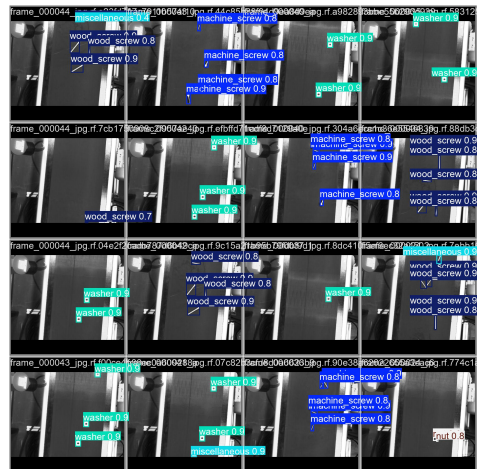


Figure: Prediction: yolov8n

Results

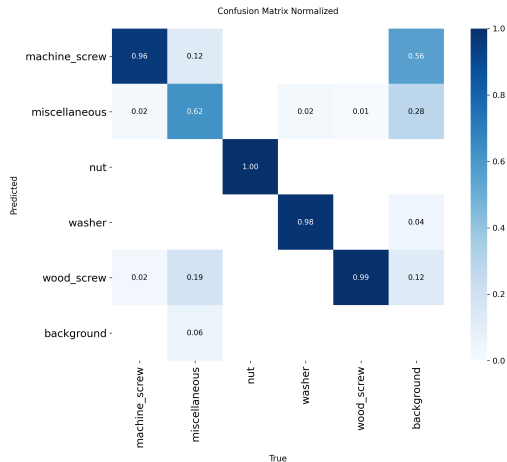


Figure: Normalized Confusion Matrix: yolov8n

References

- ▶ Slide 3 (picture): <https://www.raspberrypi.com/products/raspberry-pi-5/>
- ▶ Slide 5 (picture & specifications):
<https://en.ids-imaging.com/download-details/AB00606.html>
- ▶ Slide 6 (info & color palette): <https://roboflow.com/brand#colors>
- ▶ Slide 7 (picture): <https://www.raspberrypi.com/products/ai-hat/>

References

- ▶ Redmon, Joseph, et al. "You only look once: Unified, real-time object detection." *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2016.
- ▶ Lin, Tsung-Yi, et al. "Microsoft coco: Common objects in context." *European conference on computer vision*. Cham: Springer International Publishing. 2014.